

2.7 PARALLEL IMAGING: ASSET, ARC & ACCELERATION

SHORTER SCANS WITH SMARTER DATA COLLECTION

Parallel imaging uses multiple coil elements and their unique sensitivity profiles to undersample k-space, then mathematically reconstruct the missing data. Result: shorter scan times with a trade-off in SNR.

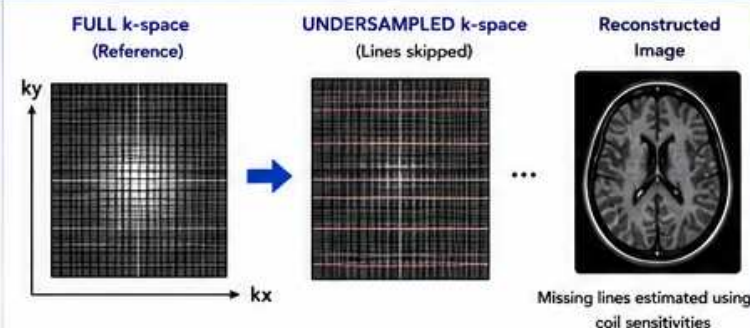


KEY CONCEPT

We intentionally skip some k-space lines. Coil sensitivity fills in the gaps. Higher acceleration = faster scans, but more potential noise and artifacts.

↑ Increase / Improve
↓ Decrease / Reduce
— No / Minimal Effect

1 THE IDEA BEHIND PARALLEL IMAGING



We collect fewer phase-encode lines → less data → faster scan. Coil sensitivity information allows us to rebuild the full image.

2 COMMON PARALLEL IMAGING METHODS

Method	How It Works (Simplified)	Where Used
ASSET (GE)	Uses multiple coil elements (spatial sensitivity) to solve for missing k-space.	Most GE systems
ARC (GE)	Uses a calibration region + coil sensitivities (similar to GRAPPA) to reconstruct.	GE systems (newer) replacing ASSET
GRAPPA (General)	Calibration data used to synthesize missing lines.	All vendors (basis of many PI methods)

Different names, same goal: speed without losing all the detail.

3 ACCELERATION FACTOR (R)

R = Acceleration Factor
How much we speed up the scan by undersampling.

R = 1	No acceleration (Full sampling)	100%
R = 2	Every other line skipped	50%
R = 3	Two lines skipped, one acquired	33%
R = 4	Three lines skipped, one acquired	25%

$$\text{Scan Time} \approx \frac{1}{R} \text{ (ignoring other factors)}$$

4 EFFECT OF ACCELERATION (R)

Effect	As R Increases
Scan Time	↓ Decreases (faster)
SNR	↓ Decreases ~ $1/\sqrt{R}$ (noise increases)
Artifacts	↑ More likely (PI, g-factor, aliasing)
Spatial Resolution	— No direct effect
Motion Sensitivity	↑ Increases
Distortion (EPI)	↑ May increase
Coil Element Requirement	↑ Higher R needs more coil elements

Rule of Thumb

Start with R = 2. Use R = 3–4 only when needed. Match R to coil and anatomy.

5 VISUAL EXAMPLES (SAME PATIENT, SAME SEQUENCE)



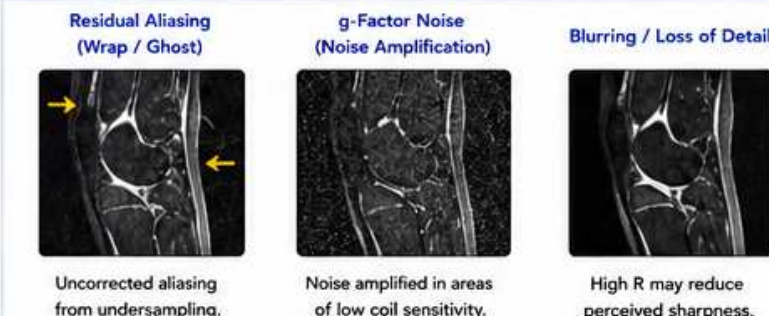
6 COIL DEPENDENCE

Coil Type (Channels)	Typical Max Effective R	Notes
8–16 ch	R = 2	Limited
20–32 ch	R = 3	Good
32+ ch	R = 3–4	Better
64+ ch	R = 4–6	Best

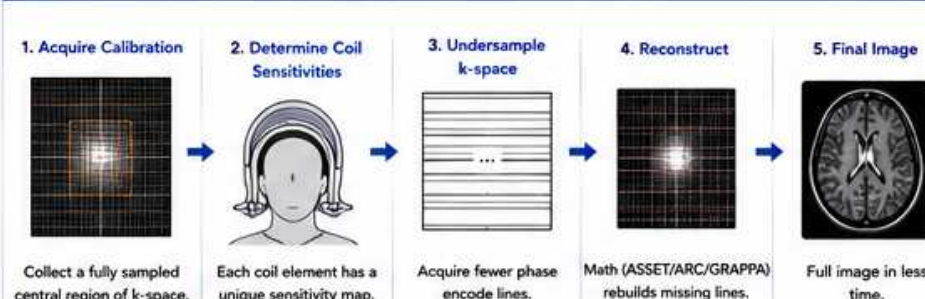
More channels = better PI performance

Insufficient coil channels → higher noise and more artifacts at the same acceleration.

7 COMMON PARALLEL IMAGING ARTIFACTS



8 WORKFLOW: HOW PI WORKS (SIMPLIFIED)



9 PRACTICAL GUIDELINES

- ✓ Use the highest R that maintains diagnostic quality.
- ✓ For routine studies, R = 2 is the sweet spot.
- ✓ Higher R is useful for breath-hold, cardiac, and high-resolution 3D scans.
- ✓ Check for artifacts when increasing R.
- ✓ Ensure the correct coil is selected.
- ✓ For small FOV or off-center anatomy, PI performance may degrade.
- ✓ Always review the final image, not just the scan time.



10 PI VS. OTHER TIME-SAVING METHODS

Method	How It Saves Time	Trade-offs
Parallel Imaging (PI)	Undersamples k-space (R = 2, 3, 4...)	↓ SNR, artifacts
ETL / TSE Factor	Fewer excitations per line	↑ Blurring, artifacts
Higher Bandwidth	Shorter dwell time per sample	↓ SNR
Lower NEX	Fewer averages	↓ SNR
Smaller FOV	Fewer phase encode lines	↓ Resolution, wrap risk

PI is the most powerful tool to reduce scan time with the best balance of quality and speed.

11 ENGINEER CHECKLIST

- ✓ Is the appropriate acceleration factor (R) used?
- ✓ Is the selected coil adequate for the chosen R?
- ✓ Are parallel imaging artifacts present?
- ✓ Is the calibration region correct and centered?
- ✓ Could adjusting R improve the scan time without harming image quality?
- ✓ Have other parameters been optimized?

⚠ Too high R + low coil elements = noisy images and PI artifacts.



KEY TAKEAWAY

Parallel imaging uses coil sensitivity to undersample k-space and shorten scan time. Match the acceleration factor (R) to the coil and clinical need — more speed, but mind the trade-offs.